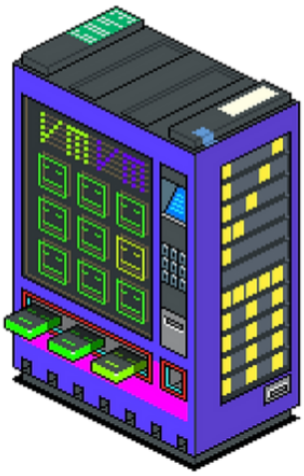
Marginal
Incentive Simulation Report



VENDING MACHINE

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1 Objective

The Marginal token system has been designed in collaboration with the core team over the past 2 months. Due to the complexity of this 'mature state' model and the time constraints regarding launch, it requires a progressive rollout.

To achieve this progressive design rollout, the model must be broken into achievable segments that make strategic sense to launch consecutively. The first two stages are critical and the final stage can be implemented at the team's discretion based upon market reception. This simulation report focuses on parameterizing the first two versions which are outlined in the CDR document.

2 Summary of Recommendations

Parameters	**Recommended Values**
LP revenue share percentage	75%
sMARG revenue share percentage	25%
Allocation of MARG supply for emission to LPs	12.5%
MARG emission period for LPs (uniform distribution)	730 days
Allocation of supply for esMARG emission to sMARG	15%
V2 esMARG emission period for sMARG (uniform distribution)	730 days

3 Assumptions

3.1 Historical Data

For the purpose of this simulation, two historic data scenarios were used to test the current system under varying liquidity and fee conditions.

There are few live protocols that function similarly to Marginal. To account for this, data was used to prioritise emulation of likely Marginal liquidity dynamics. Adjustments were then made to the data to account for likely derivative volume demand.

3.1.1 Adjusted Uniswap V3 Data

Historic data from Uniswap over the past year was used as the first scenario to mimic the permissionless nature of Marginal pool creation. In addition to the liquidity dynamics, Uniswap will also be the protocol that provides the price oracle infrastructure for Marginal pools. It also has a variable fee structure similar to Marginal pools.

To account for the significant scale of Uniswap volume and liquidity TVL, a scaling factor was used to reduce these numbers roughly in line with derivative competitors such as GMX. The scaling factor used was 0.1 to decrease the Uniswap data by a factor of 10.

3.1.2 GMX-adjusted Uniswap V3 Data

Historic Uniswap data likely emulates Marginal liquidity dynamics well, however, the demand side for Marginal is likely to be more in-line with a perpetual swap derivative platform than a spot trading DEX. In addition, the fees collected from Marginal's derivative platform include position changes (open and close), borrow rate, and liquidation fees.

In addition, to prevent unrealistically high levels of initial TVL and fees from the data influencing simulation results, both TVL and fees were time-weighted such that throughout the simulation period the TVL and fees would gradually increase to meet the comparable levels of adjusted Uniswap V3 data.

To account for the increased volume demand for derivatives, the wider fee capture of derivatives, and the growth environment experienced by new protocols in their first year, the Uniswap data was then adjusted with two further changes:

- GMX's fee:liquidity ratio during its first year of operation was used to adjust the daily Uniswap fee data to mimic derivative fee capture.
- A daily scaling factor to emulate a new protocol growth over its first year.

This created a data scenario that emulates Uniswap's permissionless market liquidity dynamics and accounts for the greater volume demand and fee capture of derivative platforms. This data scaling factor also enables us to simulate under the conditions of a new protocol experiencing growth in its first year.

The same time-weighting adjustment was applied to GMX-adjusted Uniswap V3 fee data as it was done for adjusted Uniswap V3 data.

The historical data scenarios are displayed below.

Historical Data Type	**TVL Adjustment**	**Fee Adjustment**
Adjusted Uni V3	0.1	0.1
Adjusted Uni V3	0.1	0.075
Adjusted Uni V3	0.1	0.05
GMX-adjusted Uni V3	0.1	0.1
GMX-adjusted Uni V3	0.1	0.075
GMX-adjusted Uni V3	0.1	0.05

Table 1: Historical Data Scenarios

3.2 Price Scenarios

Token price dynamics in the context of the model influence the final results significantly. Because price is an extremely variable and impossible to predict input, multiple price scenarios to account for a range of market conditions and product performance are used to test the system in most likely conditions.

The price of $MARG was modelled using Geometric Brownian Motion to emulate randomness and volatility likely to be experienced in a live market. To parameterise volatility for this simulation, historic 30-day volatility for Ethereum was used as a base range of volatile conditions. Ethereum's historic volatility ranged between 2% and 9% and this range will be used for this simulation.

Based on information from the core team, a starting token FDV of $50M will be targeted. This will be used to set the starting price for the token. To account for multiple possible scenarios, price drift values will be used to target a number of final target FDVs. The price drift scenarios for each volatility condition are displayed below.

Price Multiple at EOY	Drift Factor	FDV at EOY
1/20	-0.00820748568	$2.5M
1/10	-0.00630845231	$5M
1	0	$50M
10	0.00630845231	$500M
20	0.008207485681	$1B

Table 2: Price Drift Scenarios for Each Volatility Condition

For each price drift target, the 2 volatility settings are used for a total of 14 unique price scenario combinations.

3.3 Model Mechanisms

3.3.1 Time Interval

Each time interval in the model was assumed to be 1 day.

3.3.2 Staking Durations

The staking mechanism was modelled taking into account the upfront lock durations and the 'unlocked' staked capital. The average upfront lock capital percentage and average lock duration was taken from a combination of historic Balancer, Curve, and GMX data to find average respective lock durations. The daily staking distribution was then used in calculating the likely split of yield between staker groups.

3.3.3 Upfront Lock

The upfront locked capital was accounted for by taking an average amount of locked capital from protocols like Balancer and Curve. This average lock duration was then extrapolated using a Gaussian distribution to estimate lock times for the entirety of 'upfront locked capital'.

4 Metrics

The simulation testing was carried out in two stages for both versions. The first objective was to optimise parameters that impact the primary performance metric of the model. Upon satisfaction of these benchmarks under a range of likely conditions, the simulations can focus on granular details such as lock capital distribution and unstaking slashing rates.

4.1 Primary Metrics

The primary metrics backtested for in this report were:

- LP APR (averaged across pools) - V1 and V2
- sMARG (staked MARG) APR - V2

4.2 Secondary Metrics

The secondary metrics explored in this report were:

- sMARG multiplier points calculation
- Yield distribution for sMARG users
- V1 and V2 reward distributions

4.3 Primary Metrics

APR was chosen as the primary metric for this validation process as it best indicates the competitiveness of incentives under different scenarios of Marginal protocol.

APR is calculated as the APR given daily rewards and the underlying liquidity. APRs across the backtesting scenarios will be validated against the benchmark APR ranges.

APRs for both LPs and sMARG holders are made up of two components:

- Share of fees generated from the protocol
- MARG / esMARG emission

4.3.1 LP APR

To find historic APR benchmark ranges for Marginal protocol, a combination of historic data was used. Based on agreement with the team, an APR at least on par with the corresponding Uniswap V3 liquidity pool for a Marginal market is the minimum target.

Benchmark Ranges for LP APR	**Lower Bound**	**Median**	**Upper Bound**
Fee APR	15.63%	26.04%	42.32%
MARG APR	7.00%	19.65%	42.97%
Total APR	22.63%	45.69%	85.29%

Table 3: Benchmark Ranges for LP APR

4.3.2 sMARG APR

The benchmarks for sMARG APR come from the gold standard of derivative-based native token revenue share, GMX. No other on-chain derivative platform that has a similar native token revenue share component has as rich data as GMX. For this reason, 1st year staked GMX APRs were used to benchmark Marginal's APR performance.

Benchmark Ranges for sMARG APR	**Lower Bound**	**Mean**	**Upper Bound**
Fee APR	16.63%	31.49%	37.79%
MARG APR	46.03%	64.27%	76.83%
Total APR	69.20%	95.76%	113.45%

Table 4: Benchmark Ranges for sMARG APR

4.4 Secondary Metrics

4.4.1 Total Revenue Share per Point

Points during V1 of the Marginal system roll-out represent a relative share over two components of reward:

- The total revenue share earned during that period
- The esMARG allocation during that period

The total amount of reward/point earned is purely an observation metric. It is important, however, that the relative distribution is effective. The rate at which an individual MARG staker earns points for MARG staked is calculated as:

$$P_{V1} = (\mu M_s \cdot t_d) \cdot \left(1 + \frac{\mu MP \cdot 0.5}{\mu M_s}\right)$$

Where:

- P_{V1} = V1 points
- μM_s = Average staked MARG
- t_d = Time in days
- μMP = Average multiplier points

4.4.2 Revenue per Point Rate

The revenue per point rate is calculated as:

$$R_p = \frac{\Sigma R}{\Sigma P}$$

Where:

- R_p = \$ revenue per point
- ΣR = Sum of revenue allocated to staked MARG over V1 period
- ΣP = Sum of points earned over V1 period

4.4.3 esMARG per Point

The esMARG per point rate is calculated as:

$$M_{es} = \frac{\Sigma M_{es}}{\Sigma P}$$

Where:

- M_{es} = Individual user esMARG distributed in V1
- ΣM_{es} = Sum of esMARG allocated to staked MARG over V1 period
- ΣP = Sum of points earned over V1 period

At the crossover to V2, points earned will be converted into esMARG for users who staked their MARG tokens during V1.

4.4.4 Points Earned for sMARG

As an example, a user that staked 100,000 MARG tokens throughout the V1 period will:

- Receive 20,355 esMARG
- Receive \$132.825 in fees

5 Parameter Recommendations

Through the back-testing across various scenarios with diverse parameter sets, the parameter values for the launch of V1 and V2 have been suggested and are outlined in the following table:

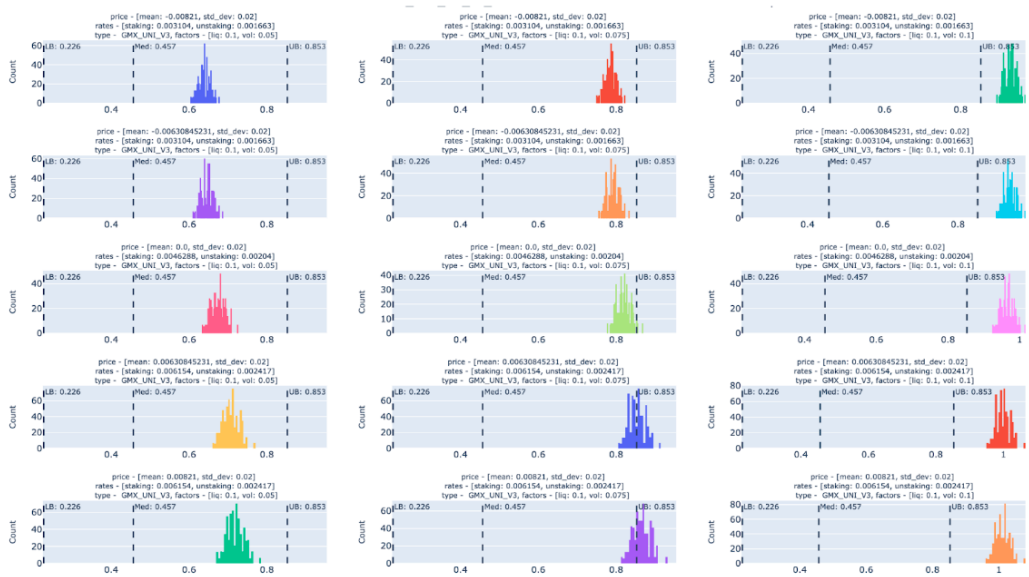
Parameters	**Recommended Values**
LP revenue share percentage	■
sMARG revenue share percentage	25%
Allocation of MARG supply for emission to LPs	12.5%
MARG emission period for LPs (uniform distribution)	730 days
Allocation of supply for esMARG emission to sMARG	15%
V2 esMARG emission period for sMARG (uniform distribution)	730 days

Table 5: Recommended parameter values for the launch of V1 and V2

A prudent approach has been taken in arriving at the recommended values for the parameters, in order to ensure that the token system satisfies the benchmark ranges in the adverse scenarios that have been simulated over. Therefore, the results from the favourable scenarios may be expected to outperform the benchmark ranges.

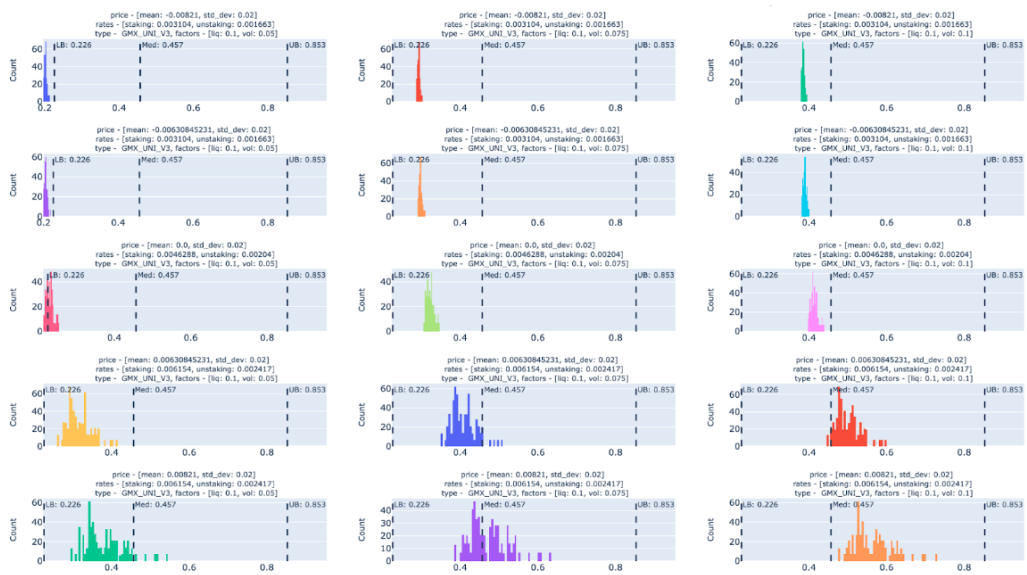
The graphs below show the distribution of APRs for LP and sMARG groups in V2 under different historical data and price scenarios using the parameters recommended above.

5.1 Distribution of LP APRs in V1 using GMX-adjusted Uniswap V3 data



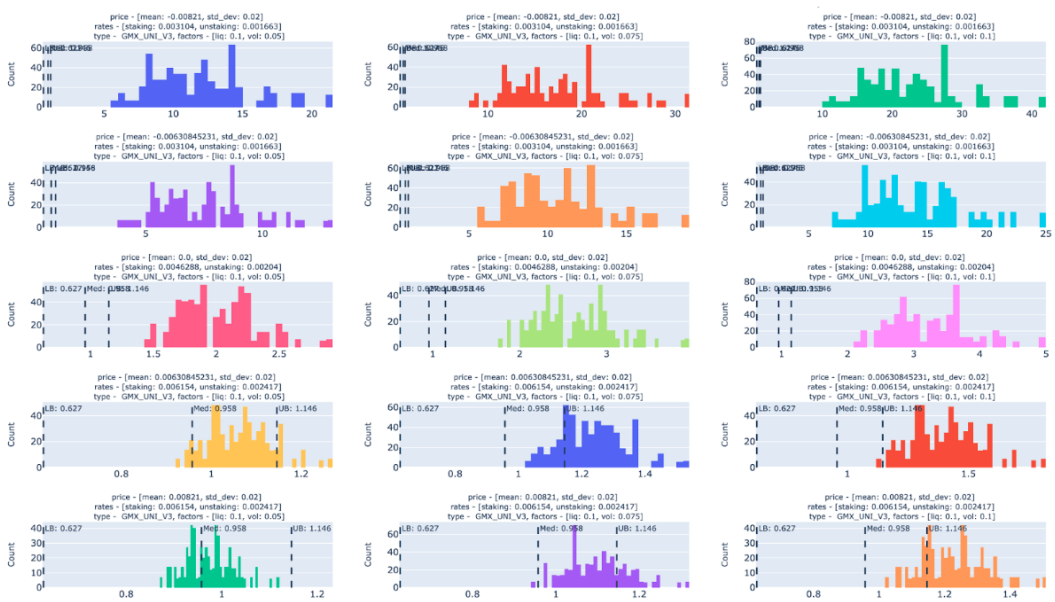
Plots above show the distribution of average LP APRs in V1 from the simulation runs using GMX-adjusted Uniswap V3 (for fees) data for given price drift and volatility, as well as the benchmark ranges for LP APRs.

5.2 Distribution of LP APRs in V2 using GMX-adjusted Uniswap V3 data



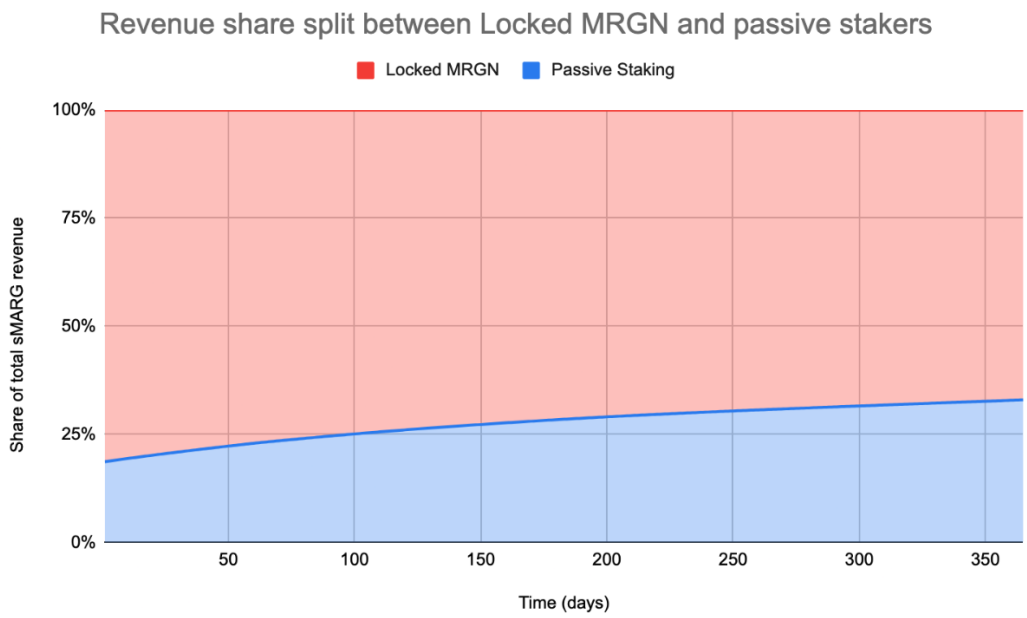
Plots above show the distribution of average LP APRs in V2 from the simulation runs using GMX-adjusted Uniswap V3 (for fees) data for given price drift and volatility, as well as the benchmark ranges for LP APRs.

5.3 Distribution of sMARG APRs in V2 using GMX-adjusted Uniswap V3 data



Plots above show the distribution of average sMARG APRs in V2 from the simulation runs using GMX-adjusted Uniswap V3 (for fees) data for given price drift and volatility, as well as the benchmark ranges for sMARG APRs.

5.4 sMARG Yield Distribution



The chart above displays the estimate absolute real yield and esMARG distribution between passive stakers and upfront locked capital.

The above chart was built to demonstrate the yield distribution between the two categories of sMARG stakers. It takes the output of the simulation from the run with a 10x price multiple, 0% fee reduction with the final parameters and it assumes a 60:40 split between upfront lockers and passive stakers respectively. It takes the average lock time of Curve and Balancer (95.5%) and applies that to the year lock duration here on a daily basis to generate multiplier points. The passive stakers earn multiplier points at a rate of 200% APR. The revenue share proportion is then calculated based on the addition of sMARG and MPs for each staker type.

6 Sensitivity Analysis

This table summarises the impact on the LP and sMARG APRs following the changes in environmental variables, to highlight the behaviour of the model outside of the ranges that have been simulated over.

Environmental Factors		Effect on sMARG APR	Effect on LP APR
Liquidity	Increase	Positive 2nd order effect	Negative 1st order effect
	Decrease	Negative 2nd order effect	Positive 1st order effect
Fees	Increase	Positive 1st order effect	Positive 1st order effect
	Decrease	Negative 1st order effect	Negative 1st order effect
MARG Price Drift	Increase	Negative 1st order effect	Positive 1st order effect
	Decrease	Positive 1st order effect	Negative 1st order effect
MARG Price Volatility	Increase	More extreme distribution	More extreme distribution
	Decrease	Less extreme distribution	Less extreme distribution
Net Staking of MARG	Increase	Negative 1st order effect	No Effect
	Decrease	Positive 1st order effect	No Effect

Table 6: Impact on LP and sMARG APRs from changes in environmental variables

6.1 Liquidity

Lower capital utilisation of the protocol (i.e. ratio of volume to liquidity) via greater liquidity provision negatively impacts LP APR in two ways:

- Greater dilution of fee revenue share amongst a larger liquidity base.
- Lower value of MARG rewards relative to a larger liquidity base.

Changes in levels of liquidity provision do not impact sMARG APR directly, however, it is anticipated to have second-order effects - for example, reduced liquidity provision leading to reduced fee revenue generated from protocol due to higher slippage. Lower level of liquidity provision for a given level of trade volume will increase the LP APR via both MARG and fee yield components whilst all sMARG APR remains unchanged.

6.2 Fees

Higher capital utilisation rate achieved via increased fees relative to liquidity will increase the fee yield component of LP. Higher capital utilisation rate will also increase the fee yield component for sMARG. Lower capital utilisation rate via lower fees will have opposite effects to those described above.

6.3 MARG Price Drift

Positive price drift of MARG token will have a positive impact on LP APR, via higher MARG reward amounts relative to the LP liquidity values, whilst the fee yield component remains unchanged. However, it is expected to have a negative impact on sMARG APR via the fee yield component, as the base value of sMARG is increased via the increasing MARG price. Negative price drift will have the opposite effect on LP and sMARG APRs, as well as genesis LP APRs and MARG token inflation rate.

6.4 MARG Price Volatility

Price volatility impacts the kurtosis (i.e. tailedness) of the distribution of APRs for the LP and sMARG tokens. Changes in price volatility impact the same components of the LP and sMARG that price drift does:

- MARG component of LP APR
- Fee and MARG components of sMARG APR

6.5 Net Staking of MARG

Greater net staking is achieved via either a higher staking rate of MARG tokens or a lower unstaking rate of sMARG tokens, or both. Greater net staking reduces sMARG APR - total return is diluted as both fee and MARG yield components are now shared amongst a larger sMARG token base. Smaller net staking has the opposite effects to those described above. The effect of changes in net staking of MARG tokens will impact LP APR indirectly via changes in MARG price via the portion of circulating supply staked - this effect has not been quantified in this modelling exercise.

7 Backtesting Findings

The simulations employed backtesting across a spectrum of parameters, including LP revenue share percentages, different levels of MARG emissions as LP incentives, esMARG emissions to sMARG, against a backdrop of historical data and price scenarios. This rigorous approach allowed us to evaluate the system's performance under conditions ranging from different price drifts, volatilities, and varying degrees of fee per TVL reduction.

The backtesting process entailed a detailed analysis of metric distributions for both V1 and V2 phases, across varying parameter sets.

7.1 Protocol Fees

The simulations revealed that a fee per TVL comparable to Uniswap V3's does yield adequate trading revenue to satisfy LP APR benchmarks.

Adjusting fee per TVL based on GMX for Uniswap V3 data produced distributions that comfortably outperformed the median benchmarks in adverse fee per TVL scenarios (50% fee reduction from normalised data).

Distribution of LP APRs in V1 under the Uniswap data is shown in Appendix A.

7.2 Revenue Share

Different revenue share splits tested:

	Target LP Fee APR	Required fee per TVL (daily)
Lower Bound	15.63%	$0.000571
Median	26.04%	$0.000951
Upper Bound	42.32%	$0.001546

Table 7: Minimum fee per TVL required to meet LP fee APR benchmarks, given 75% revenue share to LPs

- 80% LP vs 20% sMARG
- 75% LP vs 25% sMARG
- 70% LP vs 30% sMARG

Across different configurations, Total LP APR meets the minimum benchmarks using 75% revenue share for LPs. Scenarios with 25% fee reduction and no fee reduction showed in meeting and approaching the median APR benchmarks.

The other two revenue share splits resulted in either one of LP fee APRs or sMARG fee APRs not meeting the benchmark ranges in the historical data and price scenarios simulated over.

7.3 MARG Emission Rates

Allocation of MARG for emission to LPs and sMARG both started with 10% in the simulation process and were gradually increased in 0.5% intervals until the results showed that the resultant APRs were satisfactorily in the benchmark APR ranges for LP and sMARG respectively.

The shape of emission tested during the simulation process was a uniform distribution of the allocation over the emission period.

Given the requirement to meet the benchmark APRs for MARG component for LPs and sMARG, different values for the following parameters were simulated over:

- Allocation of supply for MARG emission to LPs
- Allocation of supply for esMARG emission to sMARG

The below values were obtained via the simulation process by taking into account two counteractive objectives - ensuring that the benchmark APRs were satisfied in relevant scenarios whilst not over-emitting MARG tokens, which could create unwarranted sell-pressure.

Given the fact that the benchmark ranges for MARG APRs were obtained from historical GMX data from launch which was in a positive price drift environment, it was anticipated that the MARG APRs would not comfortably meet the benchmark ranges in other price scenarios.

The negative MARG price drift scenarios are likely to be counteracted by the high APRs for sMARG (due to low MARG price) in such scenarios, which will increase demand to buy and stake MARG, creating buy pressure for MARG tokens.

The subsidisation from fee APR allows the total APRs for LPs and sMARG to meet the benchmark requirements, even when MARG APR themselves fall short of the benchmark APRs in zero and negative price drift scenarios.

To align with benchmark APRs for the MARG component for both LPs and sMARG, simulations were conducted on different allocation percentages for MARG emissions to LPs and for esMARG emissions to sMARG.

The finalised allocations were:

- 12.5% for MARG emission to LPs over 730 days
- 15% for esMARG emissions to sMARG over 730 days (to achieve at least the median benchmark MARG APR of 64%).

These allocations balanced the objectives of meeting APR benchmarks in various scenarios while avoiding excessive MARG emission, which could lead to undesirable sell pressure.

Despite challenges in matching historical GMX benchmark ranges under different price conditions, the strategy anticipates that high APRs for sMARG in adverse price scenarios would bolster demand for MARG, thereby counteracting the existing negative price drift.

The emission durations are currently parameterised to provide competitive yield for both Marginal LPs and staked MARG holders over the duration of 2 years from launch date. However, during the course of this time frame and depending on protocol strategy decisions such as implementing the V3 model, it would be wise to reassess the emission rates once live data is obtained to target revenue share sustainability.

In addition to reducing emissions in favour of revenue share reward support, the launch of V3 and the gauge system requires a redistribution in the weighting of rewards. The gauge model requires consistent LP emissions. When this feature is implemented, a re-assessment of the MARG emission distribution and the potential of implementing some inflation toward the gauge system should be considered.

Distribution of LP's MARG APRs is V2 are shown in Appendix B.

7.4 Strategic Considerations

The findings underscore the importance of fine-tuning fee structures and allocations for MARG to ensure competitive APRs while maintaining a balanced and sustainable ecosystem.

The simulation testing suggests that the proposed Marginal token system, given the recommended parameters, is capable of counteracting adverse price conditions for MARG token by bolstered sMARG APR.

However, the token model itself does not create an incentive to trade on the protocol and hence generate fees for revenue share, and thus sufficient trade volume and fees are expected to be the lifeblood of the Marginal token system. If, once the protocol is live, trading volumes are underperforming the set fee/TVL benchmarks listed above, it is worth considering the implementation of the proxy trading reward feature detailed in the V3 model of the CDR document.

7.5 Stress Scenarios

Stress scenarios are defined as conceivable conditions where multiple environmental factors, such as MARG price and fee per TVL, simultaneously reach unfavourable levels, unlike sensitivity analyses which examine the impact of varying a single factor across different states.

In scenarios characterised by high sell pressure on the MARG token coupled with low fee per TVL, we observed a decline in both MARG and fee APR components for LPs and sMARG holders. Simulations have consistently shown that scenarios combining negative price drift and low fee per TVL fare the worst among all tested conditions.

For sMARG holders, the detrimental effects of adverse environmental factors are mitigated by a reduction in the base value of sMARG holdings (through a declining MARG price), which paradoxically results in substantially high APR values.

Conversely, for LPs, there exists no analogous compensatory mechanism to neutralise the impact in such scenarios, underscoring the critical importance for the Marginal protocol to maintain a robust fee per TVL ratio to safeguard LP incentives within competitive thresholds.

8 Limitations

8.1 Time Period Granularity

Since the unit of measure for time in the model is 1 day, the model is effective at predicting the macro functions of the system over time. Therefore, it does not model intraday trends and microeconomic functions, such as MEV.

8.2 User Behaviour Limitations

The model uses historical GMX and Uniswap V3 data to perform the backtesting and assumes no further second-order behaviour changes to the results of the Marginal token system.

8.3 Interaction Between Protocol Metrics and MARG Price

Backtesting was performed under specific scenarios where both protocol metrics like liquidity and volume and MARG price were in an adverse environment, however, the model does not include a component that outlines the impact of protocol metrics on MARG price, or vice versa.

8.4 Second-Order Impacts of Circulating Supply of MARG on MARG Price

The model does not have a relationship defined between the circulating supply of MARG tokens and the price of MARG tokens. The reason for not including this was to prevent feeding in spurious assumptions into the model, as the relationship may not be in the same direction under all circumstances.

8.5 Smart Contract Risk

The model does not assume any smart contract risk or any events that occur due to the presence of it, therefore the actual economic metrics from the model may deviate from the simulated results should such cases arise.

9 Disclaimer

This documentation is provided by Vending Machine, a consultancy firm, for informational purposes only. Vending Machine makes no representation or warranty, express or implied, for the information contained in this documentation.

The comments in this document are made in relation to the currently proposed system design by the core team. Therefore the information provided is only relevant given the current circumstances of the protocol and may become outdated should the protocol design materially change.

This simulation report has been prepared to provide an overview of the potential impacts and outcomes associated with the implementation of the Marginal token system, based on a set of assumptions and limitations that have been clearly outlined within the document. While we have endeavoured to ensure that our analysis is robust and reflective of possible scenarios, it is important to acknowledge that the simulation results may not precisely predict future in-market adoption or usage patterns of the Marginal protocol.

The models used in this report are simplifications of the real-world environment and are based on theoretical data, assumptions, and projections that may not fully account for the complex, dynamic nature of market behaviours and technological evolution. As such, stakeholders should consider the findings within this report as one of several tools for decision-making and not as a definitive prediction of future events.

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Appendices

A LP's total APR distribution in V1 using adjusted Uniswap V3 data

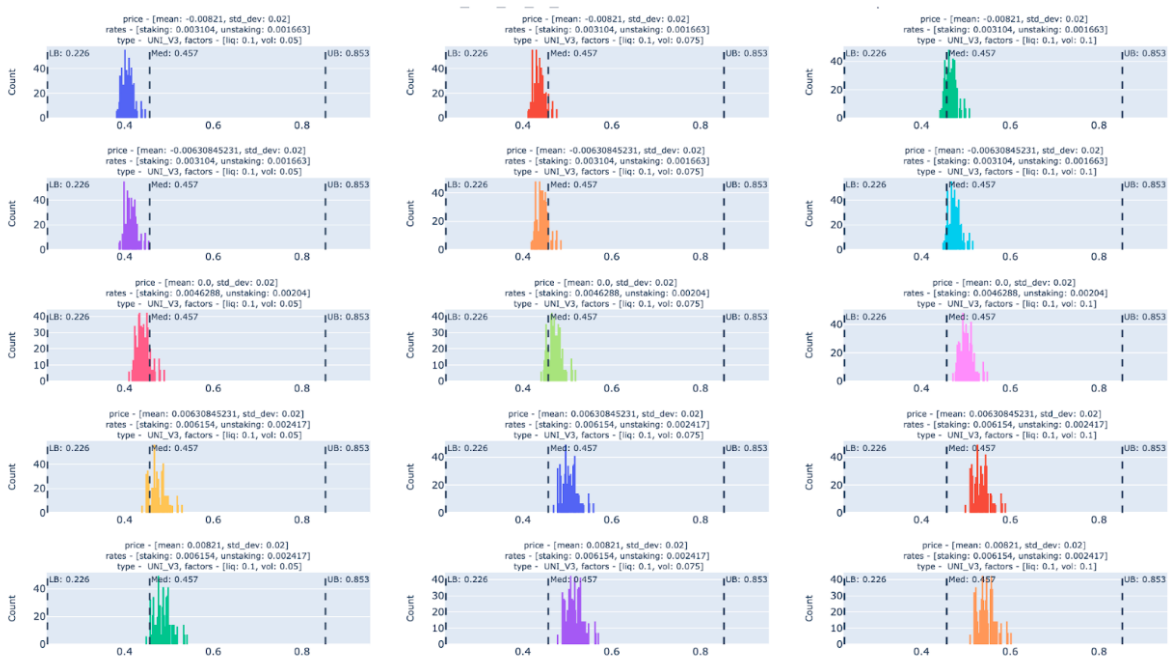

Figure 1: Plots above show the distribution of average LP APRs in V1 from the simulation runs using adjusted Uniswap V3 data for given price drift and volatility, as well as the benchmark ranges for LP APRs.

B LP's MARG APR distribution in V2 using GMX-adjusted Uniswap V3 data

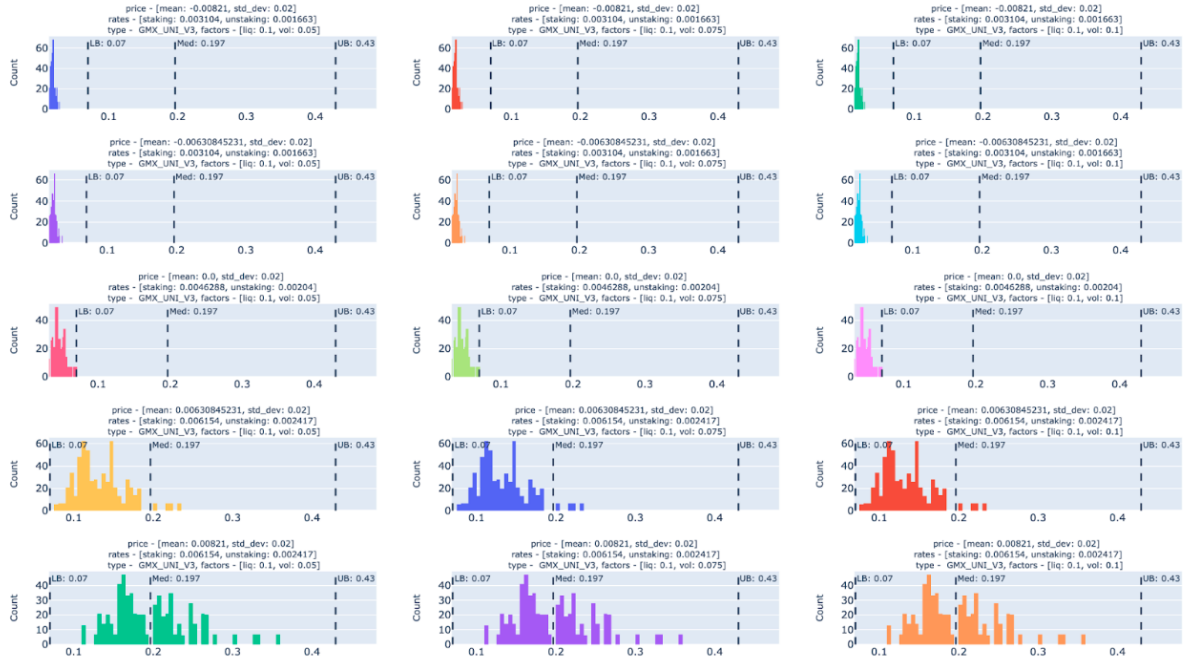

Figure 2: Plots above show the distribution of average LP MARG APRs in V2 from the simulation runs using GMX-adjusted Uniswap V3 (for fees) data for given price drift and volatility, as well as the benchmark ranges for LP MARG APRs.

C Time series of APRs using GMX-adjusted Uniswap V3 data

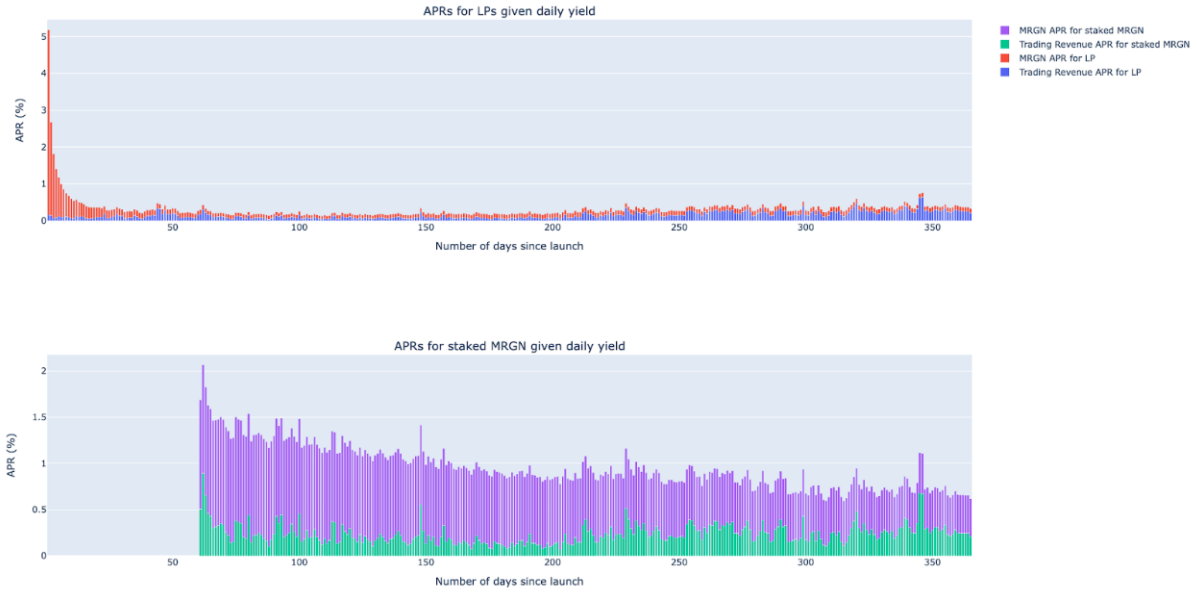

Figure 3: Plots above show the time series of LP and sMARG APRs throughout the simulation period for a 0% fee reduction form GMX-adjusted Uniswap V3 data and price drift for 10x price multiple by simulation end date.